

# Exact regions and low-polar-rank recursion in the third support spectrum of binary second-order Reed–Muller codes

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## Abstract

Let  $\mathcal{S}_n^{(3)}$  be the set of support sizes of three-dimensional subcodes of the binary second-order Reed–Muller code  $RM_2(2, n)$ . We determine several exact all-parameter regions of this spectrum. First, for every  $n \geq 4$ , the minimum value  $7 \cdot 2^{n-4}$  is isolated and the next possible support is at least  $2^{n-1}$ . Second, for every  $n \geq 7$ , a nonfull support is at most  $2^n - 2^{n-6}$ , and this bound is sharp. We determine  $\mathcal{S}_n^{(3)}$  exactly for  $2 \leq n \leq 9$ , and for every  $n \geq 8$  we classify the complete near-full interval corresponding to a nonempty common zero fibre smaller than  $2^{n-5}$ . We also prove that the support  $493 \cdot 2^{n-9}$  is impossible for every  $n \geq 12$ .

The structural part of the paper gives a finite sound-and-complete recursion whenever one nonzero output combination has nonzero zero-frequency Walsh coefficient and polar rank two or four. The recursion uses the canonical decomposition of a binary alternating pencil, exact affine mask and sign transitions for all Kronecker, degenerate regular, and nondegenerate regular blocks, and an explicit independence filter. It is constructive: every accepted state can be traced back to a three-dimensional quadratic subcode. We do not determine the complete third support spectrum; the Walsh-zero low-rank layer and the layer in which every nonzero Walsh component has polar rank at least six remain outside the theorem.

## 1 Introduction

For a linear code  $C \subseteq \mathbb{F}_2^N$ , the support of a subcode  $D \leq C$  is the union of the supports of its codewords. The generalized Hamming weights retain only the smallest support in each subcode dimension. The corresponding higher support spectrum asks for every attainable support size, and is usually much less understood. Even for Reed–Muller codes, complete higher spectra are known only in restricted parameter ranges; see Ghorpade, Johnsen, Ludhani, and Pratihari [2] for recent results and a survey of this distinction.

We study three-dimensional subcodes of

$$RM_2(2, n) = \{\text{Boolean functions on } \mathbb{F}_2^n \text{ of degree at most two}\}.$$

Write

$$\mathcal{S}_n^{(3)} = \{|\text{Supp}(D)| : D \leq RM_2(2, n), \dim D = 3\}.$$

The analogous two-dimensional support-position problem has an all-dimensional classification [1]. Passing from a quadratic pencil to a three-parameter quadratic net changes the structure substantially. Seven nonzero codewords must be compatible on a Fano parameter plane, while their affine Walsh data come from one common three-dimensional system. This prevents the ordinary quadratic weight spectrum from being used independently in each of the seven directions.

The present paper does not solve the complete three-dimensional spectrum. Instead, it closes five natural and mutually reinforcing parts of it.

1. Two sharp all-dimensional gaps are determined at the minimum and at full support.
2. The exact spectra are determined for  $2 \leq n \leq 9$ .
3. The entire near-full interval below twice the degree-six minimum weight is classified for every  $n \geq 8$ .
4. A new infinite forbidden ray is proved:  $493 \cdot 2^{n-9} \notin \mathcal{S}_n^{(3)}$  for  $n \geq 12$ .
5. All systems having a nonzero-Walsh direction of polar rank two or four are described by a finite complete canonical-pencil recursion.

The first three parts combine classical Reed–Muller weight restrictions with Fano-plane rank geometry and explicit slice constructions. The last two require affine information on alternating pencils, rather than only their polar ranks. The canonical classification itself is classical [5, 6]. Our contribution is an exact quotient of every canonical block into the masks, sign functions, and relation kernels needed for a common zero fibre, valid for arbitrary block length and compatible with constructive dynamic programming.

The proofs mix ordinary arguments with finite exact certificates. We separate their roles explicitly in Section 8. In particular, a finite computation is never used to extrapolate a theorem to unbounded  $n$ .

## 2 Quadratic systems, support, and Walsh data

Let

$$F = (f_1, f_2, f_3) : \mathbb{F}_2^n \longrightarrow \mathbb{F}_2^3$$

have linearly independent components of degree at most two, and put

$$Z(F) = |F^{-1}(0)|.$$

The corresponding subcode  $D = \langle f_1, f_2, f_3 \rangle$  satisfies

$$|\text{Supp}(D)| = 2^n - Z(F). \tag{2.1}$$

For  $a \in \mathbb{F}_2^3 \setminus \{0\}$ , set  $f_a = a \cdot F$ , and write

$$W_a = \sum_{x \in \mathbb{F}_2^n} (-1)^{f_a(x)}.$$

Pointwise, each nonzero value of  $F(x)$  is detected by exactly four of the seven functions  $f_a$ . Consequently

$$|\text{Supp}(D)| = \frac{1}{4} \sum_{a \neq 0} \text{wt}(f_a) = 7 \cdot 2^{n-3} - \frac{1}{8} \sum_{a \neq 0} W_a, \quad (2.2)$$

and Fourier inversion gives

$$8Z(F) = 2^n + \sum_{a \neq 0} W_a. \quad (2.3)$$

The polar form of a quadratic function  $f$  is

$$B_f(x, y) = f(x + y) + f(x) + f(y) + f(0).$$

It is alternating. If its rank is  $r$ , then  $r$  is even and

$$W_f(0) \in \{0, \pm 2^{n-r/2}\}. \quad (2.4)$$

The coefficient is zero precisely when the affine linear part does not vanish on the radical. If it is nonzero, its sign is the value of the induced nondegenerate quadratic form after completing the square.

Equation (2.4) yields the divisibility

$$2^{\max(0, \lceil (n-6)/2 \rceil)} \mid Z(F). \quad (2.5)$$

Indeed, in even dimension  $n = 2m$ , every nonzero Walsh coefficient is divisible by  $2^m$ ; in odd dimension  $n = 2m+1$ , it is divisible by  $2^{m+1}$ . Apply this to (2.3), including the small-dimensional boundary separately.

We use two classical facts about Reed–Muller weights.

**Fact 2.1** (minimum and next weight). *A nonzero Boolean polynomial of degree at most  $r$  has weight at least  $2^{n-r}$ . For  $2 \leq r \leq n-2$ , a weight strictly between  $2^{n-r}$  and  $3 \cdot 2^{n-r-1}$  does not occur.*

**Fact 2.2** (weights below twice the minimum). *If  $p \in RM_2(r, n)$  has*

$$0 < \text{wt}(p) < 2^{n-r+1},$$

then

$$\text{wt}(p) = 2^{n-r+1} - 2^i$$

for some integer  $i$ . This is the low-weight theorem of Berlekamp and Sloane [3], with its stated boundary included.

For the upper edge we use Warning’s second theorem in its fibre form: a nonempty common zero set of polynomials whose degree sum is  $d < n$  has at least  $2^{n-d}$  points [4].

### 3 Two all-dimensional gaps and the exact spectra through nine variables

#### 3.1 The isolated minimum

**Theorem 3.1.** *For every  $n \geq 4$  and every three-dimensional  $D \leq RM_2(2, n)$ ,*

$$|\text{Supp}(D)| = 7 \cdot 2^{n-4} \quad \text{or} \quad |\text{Supp}(D)| \geq 2^{n-1}. \quad (3.1)$$

Both values are attained.

*Proof.* A nonzero word of  $RM_2(2, n)$  has minimum weight  $2^{n-2}$ , and every nonminimum nonzero word has weight at least  $3 \cdot 2^{n-3}$ . If at most five of the seven nonzero words in  $D$  have minimum weight, their average in (2.2) already gives  $|\text{Supp}(D)| \geq 2^{n-1}$ .

Suppose that six words have minimum weight. Their six polar matrices have rank two. Every  $4 \times 4$  Pfaffian of the three-parameter polar net restricts to a homogeneous quadratic function on  $\mathbb{F}_2^3$ . Such a function cannot be nonzero at exactly one of the seven nonzero parameter points. Hence all Pfaffians also vanish at the seventh point, whose polar rank is therefore at most two. If the seventh word is not minimum, its weight is at least  $2^{n-1}$ , and the same average again gives the second alternative in (3.1). Otherwise all seven words are minimum and equality holds in the first alternative.

The subcode  $x_1 \langle x_2, x_3, x_4 \rangle$  attains the minimum. The subcode  $\langle x_1, x_1 x_2, x_1 x_3 \rangle$  attains  $2^{n-1}$ , after adjoining dummy variables as needed.  $\square$

#### 3.2 The last nonfull support

**Theorem 3.2.** *For every  $n \geq 7$ , if  $D \leq RM_2(2, n)$  is three-dimensional and does not have full support, then*

$$|\text{Supp}(D)| \leq 2^n - 2^{n-6}. \quad (3.2)$$

The bound and full support are both attained.

*Proof.* The complement of the support is the common zero set of three quadratic functions. If nonempty, Warning's second theorem gives at least  $2^{n-6}$  points. For equality, on six distinct variables take

$$f_1 = x_1 x_2 + x_1 + x_2, \quad f_2 = x_3 x_4 + x_3 + x_4, \quad f_3 = x_5 x_6 + x_5 + x_6.$$

Their common zero set fixes the first six variables to zero and has size  $2^{n-6}$ . A subcode containing the constant function one has full support.  $\square$

#### 3.3 Exact small spectra

We first record the rank geometry used by the finite arithmetic certificate. Identify an alternating form  $A$  with the linear map  $V \rightarrow V^*$  that it induces, and put  $\text{Ess}(A) = \text{im } A$ . For rank two this is the two-dimensional essential support; equivalently, if  $A = u \wedge v$ , then  $\text{Ess}(A) = \langle u, v \rangle$ .

**Lemma 3.3** (Fano rank rules). *Let  $T : \mathbb{F}_2^3 \rightarrow \text{Alt}(V)$  be linear, and evaluate the ranks of  $T$  at the seven nonzero parameter points.*

1. If  $T$  is not injective, one of the seven images is zero and, after one zero image is removed, every rank occurs with even multiplicity.
2. If at least four parameter points have rank two, then every image of  $T$  has rank at most four.
3. Suppose exactly three parameter points have rank two and some image has rank eight. Then  $T$  is injective, the three rank-two points form a line  $\{A, B, A + B\}$ , and all four points outside that line have rank at least six. If  $\dim V \in \{8, 9\}$ , at least two of those four points have rank eight.
4. Suppose exactly two parameter points  $A, B$  have rank two,  $T$  is injective, and an outside point  $C$  has rank eight. Then  $A + B$  has rank four and  $C, C + A, C + B$  have rank at least six.
5. If six parameter points have rank two, then the seventh has rank at most two.

*Proof.* If  $T$  has a kernel of dimension one, the three nonzero elements of its two-dimensional image each have two preimages among the seven nonzero parameters, while the nonzero kernel point maps to zero. If the kernel has dimension two, the remaining zero rank has multiplicity two and the unique nonzero image rank has multiplicity four; if  $T = 0$ , the remaining zero rank has multiplicity six. This proves part 1.

For part 2, first suppose that  $T$  is not injective. Its image has dimension at most two; image dimension at most one is immediate, while in dimension two part 1 forces two independent nonzero image forms to have rank two. Their sum has rank at most four, so all images have rank at most four. Now suppose that  $T$  is injective. Choose three independent rank-two forms  $A, B, C$  among four such points. Relative to this basis the fourth is either a pairwise sum or  $A + B + C$ . In the first case, say  $A + B$  has rank two. Rank-two alternating forms are decomposable, and the sum of two such forms is decomposable only when their essential two-planes meet. Thus the essential supports of  $A, B, C$  span at most five dimensions. In the second case, if the three essential supports were in direct sum then  $A + B + C$  would have rank six, so again their span has dimension at most five. Every image of  $T$  is supported there and therefore has even rank at most four.

For part 3, part 1 makes  $T$  injective. If the three rank-two points were not collinear, they would form a basis and all image forms would be supported on at most six dimensions, contradicting the rank-eight point. Hence they are  $A, B, A + B$ . Rank subadditivity gives  $\text{rank}(C + L) \geq 6$  for every point  $L$  on this line. When  $\dim V$  is eight or nine, let  $P(X)$  denote respectively the maximal Pfaffian or the vector of maximal principal Pfaffians. The three decomposable forms on the line have a common wedge factor, so the matching expansion has no repeated-update or mixed-update terms and gives

$$P(C + A) + P(C + B) + P(C + A + B) = P(C).$$

The right-hand side is nonzero. At least one updated form therefore also has rank eight, proving the last assertion.

For part 4, injectivity and the assumption that there are exactly two rank-two points show that  $A + B$  is nonzero and not rank two; rank subadditivity makes its rank at most four, hence exactly four. The same inequality applied to  $C = (C + A) + A = (C + B) + B$  gives the two rank-six lower bounds.

Finally, restrict any four-dimensional principal Pfaffian to the Fano parameter plane. It is a homogeneous quadratic Boolean function. Such a function cannot vanish at six nonzero points and be nonzero only at the seventh: its truth-set size is even. Thus every four-dimensional Pfaffian also vanishes at the seventh point, proving part 5.  $\square$

**Theorem 3.4.** *The exact support-position sets for  $2 \leq n \leq 9$  are*

$$\begin{aligned}\mathcal{S}_2^{(3)} &= \{3, 4\}, \\ \mathcal{S}_3^{(3)} &= \{4, 5, 6, 7, 8\}, \\ \mathcal{S}_4^{(3)} &= \{7, 8, \dots, 16\}, \\ \mathcal{S}_5^{(3)} &= \{14\} \cup \{16, 17, \dots, 32\}, \\ \mathcal{S}_6^{(3)} &= \{28, 32, 34\} \cup \{36, 37, \dots, 64\}, \\ \mathcal{S}_7^{(3)} &= \{56, 64, 68\} \cup \{72, 74, \dots, 128\},\end{aligned}\tag{3.3}$$

where the last progression contains even values only, and

$$\begin{aligned}\mathcal{S}_8^{(3)} &= \{112, 128, 136, 144, 148, 152, 156, 160, 164\} \\ &\quad \cup \{166, 168, \dots, 252\} \cup \{256\}, \\ \mathcal{S}_9^{(3)} &= 2\mathcal{S}_8^{(3)}.\end{aligned}\tag{3.4}$$

Again, the progression in  $\mathcal{S}_8^{(3)}$  has step two.

We give the proof in a form that separates general exclusions from finite positive certificates.

For  $n = 2$ , a three-dimensional subcode is a hyperplane in  $\mathbb{F}_2^4$ , so its support is three or four. For  $n = 3$ , the code is the even-weight code of length eight. A full-support three-dimensional even code of each length  $4, \dots, 8$  is obtained from nonzero columns of  $\mathbb{F}_2^3$  with zero column sum and full span. For  $n = 4$ , Theorem 3.1 leaves the isolated value seven and the interval from eight to sixteen; a fixed four-variable polar net realizes every affine-orbit support in that interval.

For the higher small dimensions we use the following slice identity. If  $F, L : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^3$ , with  $F$  quadratic and  $L$  affine, then

$$H(x, t) = F(x) + tL(x) \implies |\text{Supp}(H)| = |\text{Supp}(F)| + |\text{Supp}(F + L)|.\tag{3.5}$$

Three fixed polar nets have affine-orbit support sets

$$\begin{aligned}R_4 &= \{8, 9, \dots, 16\}, \\ R_5 &= \{17\} \cup \{19, 20, \dots, 32\}, \\ R_7 &= \{78, 82, 86\} \cup \{88, 90, \dots, 124, 128\}.\end{aligned}\tag{3.6}$$

Their three polar components are linearly independent. Explicit quadratic parts for these nets and an exhaustive enumeration of all vector-valued affine offsets are included in the verification package. Thus  $R_4 + R_4 = [16, 32]$ ,  $R_5 + R_5 = \{34\} \cup [36, 64]$ , and the sumset of  $R_7$ , together with doubling from dimension seven, realizes every retained value in (3.4).

It remains to justify the exclusions. In dimensions six and seven, the only arithmetic candidates not constructed are respectively 33, 35 and 66, 70. Each would require at least four rank-two points together with a polar form of rank at least six in the same Fano parameter plane. This is impossible: if the polar map is noninjective, the image has dimension at most two and the third

nonzero image has rank at most four; if it is injective, four rank-two points force the combined support of a basis of rank-two forms into at most five dimensions.

In dimensions eight and nine, every seven-word weight multiset for an unconstructed arithmetic candidate is enumerated, with every balanced word branched over its possible even polar ranks. Lemma 3.3 reduces the candidate lists to zero except for the apparent nine-dimensional pattern

$$(2, 2, 2, 6, 6, 6, 8). \quad (3.7)$$

The three rank-two points must be collinear. If  $A, B, A + B$  are the forms on that line and  $C$  has rank eight, the vector of maximal Pfaffians satisfies

$$P(C + A) + P(C + B) + P(C + A + B) = P(C). \quad (3.8)$$

This holds in dimensions eight and nine: the rank-two line has a common wedge factor, so the Pfaffian expansion has neither repeated-update nor mixed-update terms. Since  $P(C) \neq 0$ , at least one updated form also has rank eight, contradicting (3.7). Finally, the apparent supports 254 and 508 would leave only two or four common zeros, respectively, contrary to Theorem 3.2. The verification program exhausts all seven-word quadratic-weight multisets, branches over every even polar rank compatible with a balanced word, and then applies only Lemma 3.3 and (3.8). Its finite output is therefore an arithmetic certificate rather than an unproved rank-geometry oracle. This proves (3.3)–(3.4).  $\square$

## 4 The exact near-full interval

The common-zero indicator is

$$\mathbf{1}_{F=0} = (1 + f_1)(1 + f_2)(1 + f_3), \quad (4.1)$$

a Boolean polynomial of degree at most six.

**Theorem 4.1.** *Put  $M = 2^{n-6}$ . For every  $n \geq 8$ ,*

$$0 < Z(F) < 2M$$

holds if and only if

$$Z(F) = 2M - 2^i, \quad \left\lceil \frac{n-6}{2} \right\rceil \leq i \leq n-6. \quad (4.2)$$

Equivalently, all supports strictly between  $2^n - 2^{n-5}$  and  $2^n$  are

$$2^n - \left(2^{n-5} - 2^{n-\mu-5}\right), \quad 1 \leq \mu \leq \left\lfloor \frac{n-4}{2} \right\rfloor. \quad (4.3)$$

*Proof.* Apply Fact 2.2 to (4.1). It gives  $Z(F) = 2M - 2^i$ . Divisibility (2.5) forces  $i \geq \lceil (n-6)/2 \rceil$ , while positivity and the strict upper bound force  $i \leq n-6$ .

For sufficiency, use disjoint variable blocks and set

$$g_1 = x_1x_2, \quad g_2 = x_3x_4, \quad g_3 = y_1y_2 + \cdots + y_{2\mu-1}y_{2\mu}.$$

The third form is nondegenerate of positive type and has  $2^{2\mu-1} - 2^{\mu-1}$  ones. Take  $f_j = 1 + g_j$  and add dummy variables. The three polar parts lie on disjoint blocks and are linearly independent, and

$$Z(F) = \text{wt}(g_1 g_2 g_3) = 2^{n-5} - 2^{n-\mu-5}.$$

The substitution  $i = n - \mu - 5$  gives precisely the range in (4.2).  $\square$

The strict boundary is important. The value  $Z = 2M$  is attained by systems having no minimum quadratic factor and is not classified by Theorem 4.1.

## 5 Arithmetic forcing for the 19/8 ray

We now study

$$Z(F) = 19 \cdot 2^{n-9}. \quad (5.1)$$

The proof first forces a low-polar-rank negative Walsh direction, and then uses the complete recursions of Section 6.

### 5.1 The rank-two case

Suppose a negative-Walsh rank-two direction occurs. After an output change of basis, its indicator factor has minimum weight and may be normalized to  $x_1 x_2$ . Restricting to its codimension-two affine support reduces (5.1) to the intersection of the one-sets of two quadratic functions  $a, b$  in  $m = n - 2$  variables. Fourier inversion gives

$$4|\{a = b = 1\}| = 2^m - W_a - W_b + W_{a+b}. \quad (5.2)$$

At the target (5.1), division by  $2^{m-5}$  asks for three signed powers of two whose sum is 13. Negative exponents cannot occur. Indeed, at the least negative exponent exactly two of the three nonzero terms occur; opposite signs cancel and leave the impossible single power 13, while equal signs combine one level higher and either demand 12 or 14, or reduce to zero or a single signed power. Among integer powers, the only possibilities, up to order, are

$$16 - 2 - 1, \quad 16 - 4 + 1, \quad 8 + 4 + 1. \quad (5.3)$$

Their polar-rank multisets are  $\{2, 8, 10\}$ ,  $\{2, 6, 10\}$ , and  $\{4, 6, 10\}$ . The middle case violates rank subadditivity. In each remaining case the largest rank equals the sum of the other two. Their polar images are therefore disjoint, and Fourier convolution at zero gives

$$W_{a+b}(0) = 2^{-m} W_a(0) W_b(0). \quad (5.4)$$

The signs required by (5.3) have negative product, contradicting (5.4). Thus the target cannot contain a negative-Walsh rank-two direction.



## 5.2 Forcing rank two or four

Normalize the seven Walsh coefficients by

$$c_a = \frac{W_a}{2^{n-6}}.$$

Equation (2.3) and (5.1) give  $\sum_{a \neq 0} c_a = -45$ , where every nonzero coefficient is  $\pm 2^{6-r_a/2}$ . The endpoints  $+64$  and  $-64$  are impossible: the former makes an output combination zero, and the latter makes it one and hence forces  $Z = 0$ .

Assume neither  $-32$  nor  $-16$  occurs. Nonzero fractional terms cannot occur: one fractional term cannot give an integer sum, while  $k \geq 2$  such terms give a total strictly greater than

$$-k - 8(7 - k) = -56 + 7k \geq -42.$$

Positive 16 or 32 is also impossible because the remaining six terms, even if all  $-8$ , do not reach  $-45$ . Starting from seven copies of  $-8$ , the required increase is 11. The only admissible decomposition is  $11 = 4 + 7$ , so the coefficient multiset must be

$$(-8, -8, -8, -8, -8, -4, -1). \quad (5.5)$$

Through the rank-twelve point, two of the three Fano lines have rank multiset  $(12, 6, 6)$ . Equality in rank subadditivity again forces the Walsh sign of the rank-twelve member to be the product of the two rank-six signs, contrary to all three signs being negative. We have proved:

**Lemma 5.1.** *Every system satisfying (5.1) has a negative-Walsh direction of polar rank two or four.*

Section 6 gives a complete finite recurrence for both ranks. The rank-two case has already been excluded above. The rank-four recurrence excludes every remaining state and therefore closes the ray in Section 7.

## 6 A finite complete recursion for nonzero-Walsh polar rank two or four

### 6.1 Core reduction

Choose a nonzero output combination  $f_0$  with  $W_{f_0}(0) \neq 0$  and polar rank  $2\rho$ , where  $\rho \in \{1, 2\}$ . After an affine change of variables, the factor  $q_0 = 1 + f_0$  is the pullback from  $U$  of one of the two nonsingular quadrics

$$T \subseteq U = \mathbb{F}_2^{2\rho}, \quad |T| = 2^{2\rho-1} \pm 2^{\rho-1}. \quad (6.1)$$

Thus  $|T|$  is one or three for  $\rho = 1$ , and six or ten for  $\rho = 2$ . Write the ambient space as  $U \oplus R$ ,  $m = \dim R$ . The restrictions to  $R$  of the other two polar forms constitute an alternating pencil. Their cross terms with  $U$ , affine parts, and constants vary with  $u \in U$ .

For each canonical block of the pencil, and for each of the three nonzero combinations of the two residual quadratic functions, record:

- a mask  $M_i \subseteq U$  on which the Walsh coefficient is nonzero;

- an exponent  $e_i$  giving its absolute value  $2^{e_i}$ ;
- a set  $N \subseteq M_1 \cap M_2 \cap M_3$  on which the product of the three nonzero Walsh signs is negative;
- a relation kernel  $\Lambda \leq \mathbb{F}_2^2$  recording linear relations between the two noncore residual functions.

A state is therefore

$$\Sigma = (m; e_1, e_2, e_3; M_1, M_2, M_3; N; \Lambda). \quad (6.2)$$

Orthogonal direct sum gives the exact composition law

$$\begin{aligned} m'' &= m + m', & e_i'' &= e_i + e_i', \\ M_i'' &= M_i \cap M_i', & N'' &= (N \triangle N') \cap M_1'' \cap M_2'' \cap M_3'', \\ \Lambda'' &= \Lambda \cap \Lambda'. \end{aligned} \quad (6.3)$$

## 6.2 Canonical blocks and affine-product classes

Let  $\mathcal{A}_U$  be the truth masks of affine functions on  $U$ , and let  $\mathcal{Q}_2(U)$  denote the Boolean functions on  $U$  of algebraic degree at most two. Also put

$$\mathcal{P}_r(U) = \left\{ \bigtriangleup_{j=1}^r (A_j \cap B_j) : A_j, B_j \in \mathcal{A}_U \right\}, \quad \mathcal{P}_0(U) = \{\emptyset\}. \quad (6.4)$$

The alternating-pencil canonical decomposition consists of the following blocks [5, 6].

**Singular Kronecker block  $K_k$ .** In dimension  $2k + 1$ ,

$$Q_1 = \sum_{i=0}^{k-1} x_i y_i, \quad Q_2 = \sum_{i=0}^{k-1} x_{i+1} y_i. \quad (6.5)$$

For  $k \geq 1$ , the three masks independently range over  $\mathcal{A}_U$ , all exponents are  $k + 1$ , and, on their common mask  $C$ ,

$$N = P \cap C, \quad P \in \mathcal{P}_{k-1}(U). \quad (6.6)$$

The two residual polar generators are independent. The block  $K_0$  is a one-dimensional common radical. For affine zero masks  $A, B$ , its masks are

$$(A, B, U \setminus (A \triangle B)),$$

its product sign is positive, and its relation kernel is read directly from the two affine coefficients.

**Degenerate regular chain  $D_k^{(j)}$ .** In dimension  $2k$ , one of the three pencil points is degenerate. Its mask lies in  $\mathcal{P}_1(U)$ , the other two masks are all of  $U$ , its exponent is  $k + 1$ , and the other two exponents are  $k$ . On the common mask, (6.6) again holds. Only the length-one case with an identically zero degenerate residual function contributes a nontrivial relation line.

**Nondegenerate regular part  $R_d$ .** Every component on which the three rational pencil points are nondegenerate has, after local coordinates,

$$Q_1 = x^\top y, \quad Q_2 = x^\top C y, \quad C, I + C \text{ invertible.} \quad (6.7)$$

All three masks are  $U$ , all exponents are  $d$ , and  $N \in \mathcal{P}_d(U)$ . This transfer depends only on the total half-dimension  $d$ , not on the elementary divisors of  $C$ .

The last assertion follows from the exact factorization of the product-sign control matrix. Put  $D = C^\top$  and  $A = (I + D)^{-1}$ . Then

$$M_C = \begin{pmatrix} I + A & A \\ A & D^{-1} + A \end{pmatrix} = \begin{pmatrix} D \\ I \end{pmatrix} A \begin{pmatrix} I & D^{-1} \end{pmatrix}, \quad (6.8)$$

which has rank  $d$ . The analogous control ranks for  $K_k$  and  $D_k$  are both  $k - 1$ . Explicit triangular elimination gives, for all block lengths,

$$P_K = \sum_{i=0}^{k-2} (p_i + s_{i+1}) \left( \sum_{j=i+1}^{k-1} r_j + \sum_{j=i}^{k-2} t_j \right), \quad (6.9)$$

and

$$P_D = \sum_{i=1}^{k-1} ((p_1)_i + (p_2)_{i-1}) \left( \sum_{j=0}^{i-1} (r_1)_j + \sum_{j=1}^i (r_2)_j \right). \quad (6.10)$$

The left and right factors in each sum can be specified independently of the nonzero masks. Equations (6.8)–(6.10), rather than a bounded block table, prove the arbitrary-length transition formulas.

### 6.3 Terminal output and independence

For each  $\rho \in \{1, 2\}$  and each of the two choices of  $T$ , set  $m = n - 2\rho$ . Starting with the empty state, append every canonical block transition whose accumulated dimension does not exceed  $m$ , using (6.3). Retain only block sequences whose accumulated dimension is exactly  $m$ ; these and only these states are terminal. Take the union of their outputs. There are finitely many states for fixed  $m$ , because every block parameter is bounded by its dimension.

For a terminal state and  $u \in U$ , let

$$a_i(u) = \begin{cases} 2^{e_i}, & u \in M_i, \\ 0, & u \notin M_i, \end{cases}$$

and reverse the sign of  $a_3(u)$  for  $u \in N$ . If the two pure core quadratic functions take values  $c_1(u), c_2(u)$ , the residual pair-intersection number is

$$L_u = \frac{2^m - (-1)^{c_1(u)} a_1(u) - (-1)^{c_2(u)} a_2(u) + (-1)^{c_1(u) + c_2(u)} a_3(u)}{4}. \quad (6.11)$$

Enumerate the two actual functions  $c_1, c_2 \in \mathcal{Q}_2(U)$  and sum (6.11) over  $u \in T$ . The resulting system is three-dimensional precisely when, for every nonzero  $(b_1, b_2) \in \Lambda$ ,

$$b_1 f_1^U + b_2 f_2^U \notin \{0, f_0\}. \quad (6.12)$$

This condition is essential on the three-point and ten-point quadrics; removing it admits false outputs coming from dependent generators.

**Theorem 6.1** (finite low-rank recursion). *Fix  $n$ . Among all three-dimensional systems  $F$  having a nonzero output combination with nonzero zero-frequency Walsh coefficient and polar rank two or four, the attainable values of  $Z(F)$  are exactly the values produced by the finite recursion (6.2)–(6.12). Each accepted value comes with a reconstructible quadratic witness.*

*Proof.* For necessity, normalize the selected direction and decompose the restricted alternating pencil on its radical into canonical  $K_k$ ,  $D_k^{(j)}$ , and regular primary components. Equations (6.6)–(6.10) give the exact transition of every block; (6.3) is Walsh multiplication under orthogonal sum; and (6.11) is Fourier inversion on each core slice. Hence every system reaches a terminal state and output accepted by (6.12).

For sufficiency, every block transition has explicit affine coefficients. Control-matrix factorization reconstructs the prescribed sign mask, and orthogonal sum reconstructs the terminal state. The chosen core functions produce (6.11), while (6.12) removes exactly the possible generator relations. Parent pointers, block parameters, affine-product factors, and core functions therefore reconstruct a linearly independent triple realizing every accepted value.  $\square$

The theorem deliberately requires a *nonzero* Walsh coefficient. There are infinite systems with rank profile  $(2, 2, 0, 2, 0, 0, 2)$  for which all seven zero-frequency Walsh coefficients vanish. They are a boundary control, not an input to Theorem 6.1.

## 7 An infinite forbidden support ray

**Theorem 7.1.** *For every  $n \geq 12$ ,*

$$19 \cdot 2^{n-9} \notin \{Z(F) : F \text{ has three independent quadratic components}\}.$$

Equivalently,

$$493 \cdot 2^{n-9} \notin \mathcal{S}_n^{(3)}. \quad (7.1)$$

*Proof.* Lemma 5.1 forces a negative-Walsh direction of polar rank two or four. Section 5.1 excludes the rank-two case.

For rank four, specialize Theorem 6.1. If  $m = n - 4$  and  $\delta_i$  are the three residual pencil half-ranks, the normalized target equation is

$$\sum_{i=1}^3 c_i 2^{5-\delta_i} = -116, \quad c_i \in \{-6, -4, -3, -2, -1, 0, 1, 2, 3, 4, 6\}. \quad (7.2)$$

Together with the half-rank triangle inequalities, (7.2) leaves only

$$(0, 3, 3), (0, 4, 4), (0, 5, 5), (1, 2, 3), (1, 3, 3), (1, 3, 4), (1, 4, 4). \quad (7.3)$$

Thus, apart from  $K_0$ , only  $K_1, D_1, D_2$  can occur. Their integer decompositions give nine bounded canonical cores. Multiple  $K_0$  layers stabilize after two blocks under the intersection law (6.3), so it is enough to check zero, one, and at least two common-radical blocks for each core. Substituting the exact masks and sign classes from Section 6 into (6.11) excludes (7.2) in every case. This is a finite exact consequence of the arbitrary-length block theorem, not a scan over bounded  $n$ .  $\square$

At  $n = 12$ , an independent enumeration of all 65 residual pencil exponent profiles and 27 surviving canonical decompositions gives the same exclusion. The only state surviving an artificial independent-sign relaxation requires a five-point negative mask omitted by the exact length-two  $D_2$  transfer. This provides a useful destructive control for the sign coupling in the proof.

## 8 Computational proof boundary and reproducibility

The main results use finite exact certificates at several explicitly identified interfaces.

### 8.1 General arguments independent of finite enumeration

- the support/Walsh identities (2.1)–(2.3) and divisibility (2.5);
- Theorems 3.1 and 3.2;
- the explicit constructions in Theorem 4.1;
- the signed-dyadic and rank-equality arguments in Section 5;
- canonical-pencil completeness and the arbitrary-length transition formulas (6.8)–(6.10);
- termination, soundness, completeness, and witness reconstruction in Theorem 6.1;
- the dimension-independent reduction (7.2)–(7.3).

### 8.2 Finite exact certificates

- affine-orbit support sets of the three explicit polar nets in (3.6), their polar-component independence, and the corresponding slice sumsets;
- all seven-word weight and polar-rank patterns used for the  $n = 8, 9$  exclusions, including noninjective polar maps and the maximal-Pfaffian negative control;
- exact formula checks for  $K_k, D_k, R_d$ , tested on representative block lengths and regular companion matrices;
- finite state composition, relation-kernel intersection, one/three/six/ten-point terminal output, and independence positive and negative controls;
- the nine rank-four cores and stabilized  $K_0$  layers in Theorem 7.1.

The verification runner recomputes every retained certificate using exact integer, rational, or  $\mathbb{F}_2$  arithmetic. One fixed-seed routine samples regular matrices only as a regression for the symbolic factorization (6.8); the sample is not used to infer generality. A failed search and a solver timeout are never used as nonexistence evidence.

## 9 Relation to prior work and limits

Ghorpade, Johnsen, Ludhani, and Pratihari [2] determine higher support spectra for  $RM_q(2, 2)$ , where the number of variables is fixed and the field varies. The present problem fixes  $q = 2$ , lets the number of variables vary, and studies three-dimensional subcodes. The results are adjacent but neither specializes to the other.

Berlekamp and Sloane [3] provide the ordinary low-weight restriction used as a necessary input in Theorem 4.1. Ordinary codeword weights do not supply the compatibility or sufficiency of a common three-quadratic zero fibre. The latter is provided here by explicit independent constructions.

The alternating-pencil decompositions of Plût, Fouque, and Macario-Rat [5] and Macario-Rat, Plût, and Gilbert [6] are structural inputs to Theorem 6.1. We do not claim a new canonical classification. The new part is the exact affine Walsh quotient of each block, the arbitrary-length product-sign formulas, and the complete common-zero recursion with independence filtering.

The previously published two-dimensional support-position classification [1] is a precursor, not a proof of the present results. A three-dimensional system contains seven coupled directions and requires the Fano and canonical-pencil mechanisms developed here.

The following questions remain outside the scope of this paper.

1. We do not determine the complete set  $\mathcal{S}_n^{(3)}$  for every  $n$ .
2. The low-polar-rank recursion excludes Walsh-zero selected directions; the all-high-rank layer is also open.
3. We do not count subcodes of a given support size or classify quadratic nets up to equivalence.
4. We make no claim over odd characteristic or arbitrary finite fields.
5. The current literature search found no direct coverage of the frozen theorems, but it does not establish global priority or external acceptance.

## 10 Conclusion

The third support spectrum of  $RM_2(2, n)$  is not a free seven-fold combination of ordinary quadratic weights. Nevertheless, several natural regions admit exact structure. The two spectrum edges have sharp universal gaps; all spectra through nine variables can be closed; the complete near-full interval follows from a classical low-weight theorem plus a compatible three-factor construction; and the 19/8 common-zero ray is uniformly forbidden.

The canonical-pencil recursion explains why low polar rank remains tractable. After fixing one rank-two or rank-four direction, the remaining wild-looking net becomes an alternating pencil on the radical. Full marked-module orbits are not needed: affine masks, product-sign classes, and a five-valued relation kernel form a finite sufficient quotient. The unresolved part is therefore sharply located in the Walsh-zero low-rank boundary and in nets whose every nonzero Walsh direction has polar rank at least six.

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